

Semester Project: Domain Discovery, Recommendation, and Graph Intelligence

Milestone: Feature Engineering and Dimensionality Reduction (Week 5)

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1. Overview

This milestone covers the construction of the three feature matrices used as input to all downstream models (clustering, recommendation, graph analysis), and the dimensionality reduction analysis used to select compression parameters.

Code: `src/feature_engineering.py` | Artifacts: `artifacts/` | Figures: `src/generate_figures.py`

2. Dataset Snapshot

- Processed dataset: `data/processed/steam_v1.parquet` - 480,025 interaction records
- Games with tag metadata (clustering): 69,943 titles
- Games in feature matrices: 87,806 titles (full catalog)

3. Feature Matrices Built

Matrix	Shape	Type	Content
X_numeric	87,806 x 26	Dense	Numerical catalog fields. StandardScaler applied.
X_categorical	87,806 x 5,000	Sparse CSR	Top-5,000 genre/category tokens. Binary encoding.
X_text	87,806 x 5,000	Sparse CSR	TF-IDF on name + description + tags (1-2 grams).

Numeric columns (26)

Required age, price, DLC count, Metacritic score, achievements, recommendations, user score, score rank, positive/negative counts, estimated owners, average/median playtime (forever and 2-week windows), peak CCU, pct_pos (total/recent), num_reviews (total/recent), Windows/Mac/Linux flags, release year, release month, release age in days.

4. Dimensionality Reduction Analysis

PCA on X_numeric

- Components tested: up to 26 (full rank).
- Result: k=12 components capture $\geq 90\%$ of variance.
- 12 orthogonal directions cover price tiers, engagement levels, platform availability, recency.

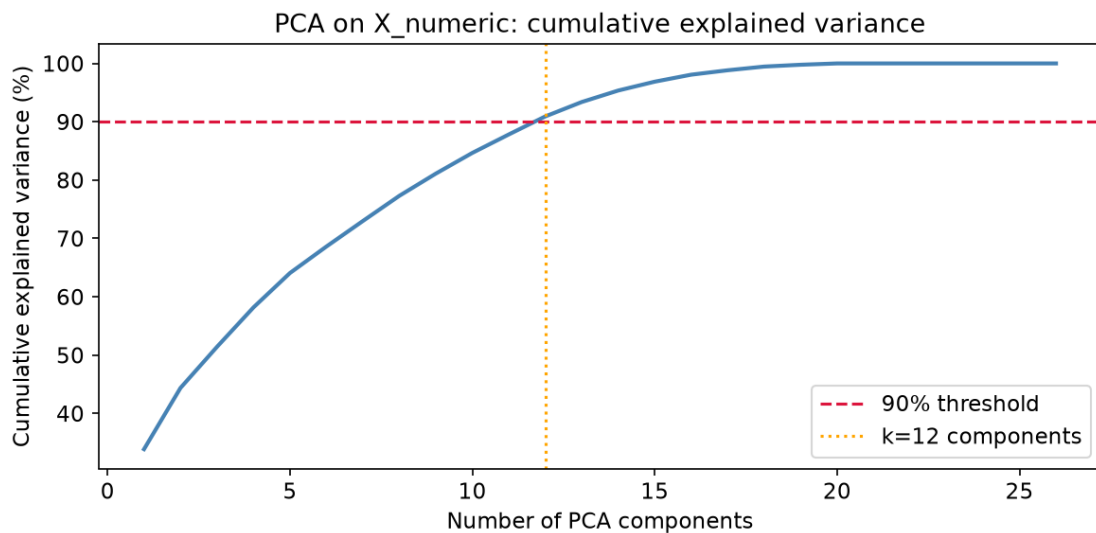


Figure: PCA on X_{numeric} : cumulative explained variance. Red dashed = 90% threshold; orange dotted = $k=12$.

TruncatedSVD on X_{text}

- Components tested: up to 300.
- Result: $k=201$ components capture $\geq 50\%$ of total squared singular value mass.
- Slow singular value decay (no sharp elbow) is typical of broad-vocabulary document collections.

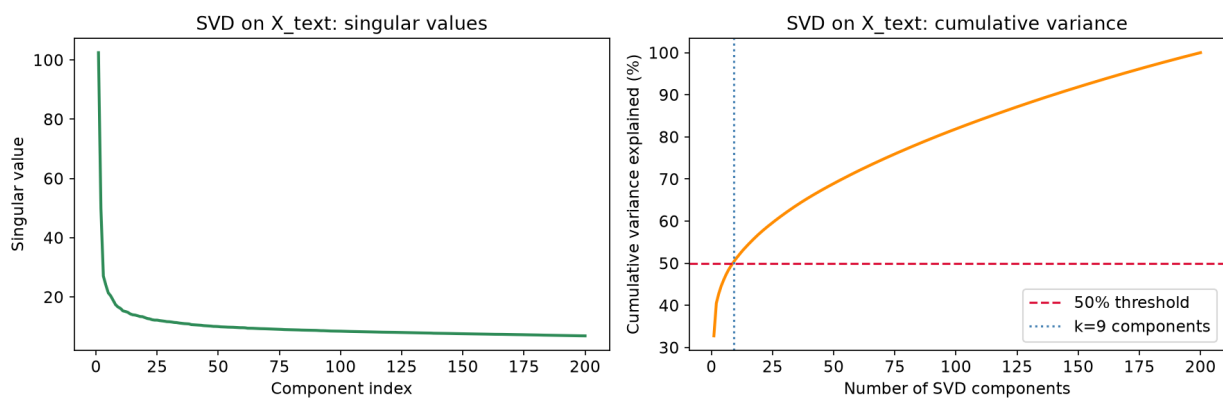


Figure: TruncatedSVD on X_{text} : singular value spectrum (left) and cumulative variance (right).

Summary

Matrix	Original dims	Reduced dims	Criterion	Method
X_{numeric}	26	12	$\geq 90\%$ variance	PCA
X_{text}	5,000	201	$\geq 50\%$ variance	TruncatedSVD
$X_{\text{categorical}}$	5,000	50 (fixed)	downstream use	TruncatedSVD

5. Feature Matrix Dimensions

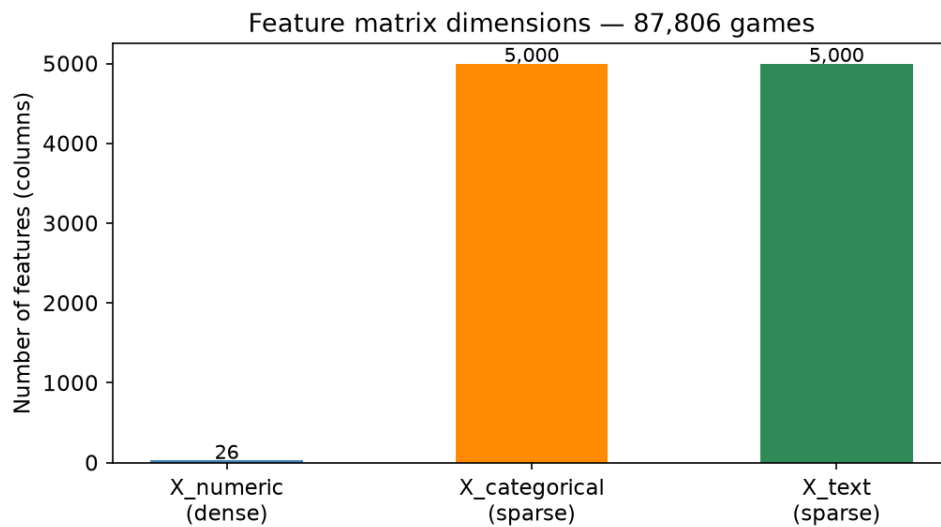


Figure: Feature column count per matrix. *X_numeric* is dense; *X_categorical* and *X_text* are sparse.

6. Reproducible Pipeline

```
# From project root  
python src/feature_engineering.py
```

Artifacts: *X_numeric.npz*, *X_categorical.npz*, *X_text.npz*, *models.pkl*, *metrics.json*

All outputs deterministic (*random_state=42*).

7. Ethics and Access Note

- Provenance: Feature matrices derived from *steam_v1.parquet*. No review text or PII.
- Authorization: CC0 license; unrestricted academic use.
- Bias note: *X_numeric* includes popularity counts introducing popularity bias (addressed in W10).